```
import torch
import pandas as pd
import numpy as np
import matplotlib.pyplot as PI t
import seaborn as sns
from datasets import load_dataset from transformers import Bert Tokenizer, BertForSequenceC1assification, Trainer, TrainingArguments from sklearn.metrics import accuracy_score, precision_recall_fscore_support, confusion_matrix

# Load dataset dataset = load_dataset ( " sms_spam " )

# Show sample records
print(dataset[ •train' ]
print(dataset[ 'train' ]

# Convert to pandas for exploration pd .
' train ' ] ) print(df.head())

# Class distribution sns .
countplot(x=df[ ' label ' ] )
plt.tit1e("C1ass Distribution") plt . show()
```

```
/usr/10ca1/1ib/python3.12/ dist-packages/ UserWarning:
The secret 'I-IF_TOKEN* does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https: //huggingface.co/settings/tokens), set it You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets. warnings .
warn(
README.md: 4.98k!? 332kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d WARNING:
huggingface_hub.uti1s._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
plain_text/train-00000-of-00001 .parquet: 100% 359k/359k 556kB/s]

Generating train split: 100% 5574/5574 78324.93 examples/s]
```

```
{ 'sms' : 'Go until jurong point, crazy. . Available only in bugis n great world la e buffet. . . Cine there got amore wat. . .\n
{ 'sms' :
'Ok lar. .
. Joking
wif u
oni. . .\n
, 'label' :
e} sms
label
0 Go until jurong point, crazy. . Available only
1 Ok lap . . . Joking wif u oni. .
2 Free entry in 2 a wkly comp to win FA Cup fina. 1 3 U dun say so early hop. . . U c already then say.
4 Nah I don't think he
goes to usf, he lives
aro. . . Class
Distribution
5000

4000

3000
```

2000

1000

1

label

Load BERT tokenizer tokenizer = Bert Tokenizer .
from_pretrained (" bert -base-uncased ")

 # Tokenization function def tokenize(batch) :

return tokenizer(batch[' sms '] , padding= 'max_ length' , truncation-True, max_ length-128)

# Apply tokenization dataset =
dataset.map(tokenize, batched=True)

Convert labels to tensor format dataset = dataset. rename_ column (" label " , " labels")

dataset . set_format(type= "torch " , columns= [' input _ ids ' ' attention mask' , ' labels '])

tokenizer_config.json: 100% 48.0/48.0 3.68kB/s]

vocab.txt: 100% 2320232k 2.25MB/s]

tokenizer.json: 100% 466k/466k 6.16MB/s]

Map: 100% 5574/5574 4341.59 examples/s]

Split dataset into train and test sets split_dataset = dataset[' train '
] . . 2, seed=42) train dataset — split_dataset['train '] test dataset =
split_dataset[' test ']

print(f"Train size: {len(train_dataset)}, Test size: {len (test_dataset) } ")

Train size: 4459, Test size: 1115

Load BERT model for classification model - BertForSequenceC1assification . from_pretrai
ned (" bert -base-uncased " , num_1abe1s=2)

config.json: 100% 570/570 53.2kB/s]

model.safetensors: 100% 440M/440M [01 15.9MB/s]

Loading weights: 100% 199/199 495.28it/s, Materializing param=bert.pooler.dense.weight]

BertForSequenceC1assification LOAD REPORT from: bert-base-uncased

Status

cls . predictions. transform. LayerNorm.weight I UNEXPECTED

cls . predictions . transform . dense . weight UNEXPECTED

cls . seq_relationship . bias UNEXPECTED

cls . predictions . bias UNEXPECTED I

cls . seq_relationship . weight UNEXPECTED

cls . predictions . transform . LayerNorm. bias UNEXPECTED I

cls . predictions . transform . dense . bias UNEXPECTED

classifier. bias MISSING

classifier. weight MISSING

Notes :

- UNEXPECTED : can be ignored when loading from different task/ architecture; not ok if you expect identical arch.
- MISSING : those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.

Define metrics def :

labels = pred . label_ ids preds = pred . predictions . argmax(-1) precision, recall, fl, - precision_recall_fscore_support(labels, preds, average= 'binary') acc = accuracy_score(labels, preds) return {"accuracy" : acc, "precision" • precision, 'recall" : recall,

```
# Training arguments training_args =
TrainingArguments( output_dir=" . /
results " , "epoch " ,
learning_rate=2e-5,
per_device_train_batch_size=16,
per_device_eva1_batch_size=16,
num_train_epochs=3,
weight_decay=0.01 , logging_dir=" . /
logs " , logging_steps=10
```

is deprecated and will be removed in vs. 2. Please set instead.

https://colab.research.google.com/drive/lfrFAoQGkumZIKBjErATYvTpCx5uehHL#scrollTo=_C5AnwEghUYm&printMode=true

+ Gemini

```
from transformers import DistilBertTokenizer, DistilBertForSequenceC1assification

# Use DistilBERT instead of BERT tokenizer = DistilBertTokenizer. from_pretrained ( " distilbert- base-uncased " ) model = DistilBertForSequenceC1assification from_pretrained("distilbert-base-uncased" , num_1abe1s=2)

# (Optional) reduce dataset size for speed small train_dataset. shuffle(seed=42) . # only 2000 ).select(range(2000)) train = test = test_dataset. shuffle(seed=42) .# only 500 select(range(500)) samples small samples

# Training arguments (lighter config)
training_args = TrainingArguments(
    output_dir="./ . / results" , "epoch
    " , eval_strategy= learning_rate=3e-5,
    per_device_train_batch_size=8, # smaller batch size
    per_device_eva1_batch_size=8, num_train_epochs=l, #
    only 1 epoch weight_decay=0.01 , logging_dir=" . / logs
    " , logging_steps=10


trainer = Trainer( model-model,
    args=training_args, train
    dataset=small train, eval
    dataset-small test,
    compute_metrics-
    compute_metrics


trainer. train()
```

Loading weights: 100% 100/100 [00:00<00:00, 367.16it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.weight

```
DistilBertForSequenceC1assification LOAD REPORT from: distilbert-base-uncased
Key                                Status

vocab_layer_norm.weight I UNEXPECTED
vocab_transform. weight
            UNEXPECTED vocab_projector.
bias        UNEXPECTED vocab transform.
bias        UNEXPECTED
vocab_layer_norm. bias
            UNEXPECTED pre_classifier.
bias        MISSING pre_classifier. weight
            MISSING classifier. bias
            MISSING classifier. weight
            MISSING

Notes :
 UNEXPECTED        : can be ignored when loading from different task/ architecture; not ok if you expect identical arch.
   MISSING : those params were newly initialized because missing from the checkpoint. Consider training on your downstream task. logging_dir* is
  deprecated and will be removed in vs. 2. Please set *TENSORBOARD LOGGING DIR* instead.
/usr/10ca1/1ib/python3.12/dist-packages/torch/uti1s/data/data10ader.py:775: UserWarning:      pin_memory' argument is set as
   true bu super() •__init_ (loader)
```

[37/250 02:45 < 16:49, 0.21 it/s, Epoch O. 14/1]

Epoch Training Loss Validation Loss